FutureStrive

Construction Intelligence Platform

Prediction Model Documentation

Crack Intelligence · Defect Volume Intelligence

This document provides a complete technical reference for both AI prediction modules in the Construction Intelligence Platform. It covers every input feature, the model used for each prediction target, dataset characteristics, and measured accuracy.

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1. Crack Intelligence Module

Predicts three outputs from 21 concrete pour parameters: (1) crack occurrence probability, (2) severity grade, and (3) crack type. Designed to flag high-risk pours before placing and drive IS 456:2000-aligned corrective action.

1.1 Dataset Overview

Property	Detail
Total Samples	5,000 rows
Feature Count	21 input features
Targets	3 (crack_occurred · crack_severity · crack_type)
Generation	Synthetic — score-based rules derived from IS 456:2000 and IS 7861
Train / Test Split	80% / 20% (4,000 train · 1,000 test)
Random Seed	42

1.2 Input Features

All 21 features are captured at pour level. Categoricals are one-hot encoded; numerics are standardised inside the sklearn pipeline.

Feature	Type	Description
concrete_grade	Categorical	Mix design grade: M25, M30, M35, M40. Higher grade = denser mix = lower permeability and crack risk.
cement_type	Categorical	OPC 43 Grade, OPC 53 Grade, PPC, or PSC. Affects hydration heat and shrinkage behaviour.
admixture_type	Categorical	No Admixture, Naphthalene-based superplasticiser, or Polycarboxylate-based superplasticiser.
water_cement_ratio_design	Numeric	Target W/C from mix design. IS 456:2000 Table 5 specifies exposure-class maxima (e.g., 0.45 for M35 moderate exposure).
water_cement_ratio_actual	Numeric	Measured W/C at site. When actual > design + 0.01 this is the strongest single crack predictor.
wc_ratio_tolerance_spec	Numeric	Maximum allowable W/C in the project quality specification.
target_slump_mm	Numeric	Workability target in mm. Linked to admixture dosage and water content.
max_aggregate_size_mm	Numeric	Nominal maximum aggregate size: 10 mm or 20 mm. Affects water demand.
planned_pour_month	Categorical	Calendar month of the pour. Captures seasonal temperature, humidity, and wind exposure patterns.
curing_method	Categorical	Ponding, Sprinkling, Wet Burlap, or Curing Compound. Affects moisture retention efficiency.
planned_curing_duration_days	Numeric	Specified curing duration in days. IS 456:2000 Section 13.5 requires 14 days minimum when mineral admixtures are used.
actual_curing_duration_days	Numeric	Actual curing achieved at site. Each day below the 14-day minimum increases crack probability.

Feature	Type	Description
spec_min_curing_days	Numeric	Project-specific minimum curing requirement (may exceed IS 456 minimum).
pre_pour_checklist_signed_off_ratio	Numeric	Fraction of pre-pour checklist items signed off (0–1). Low values indicate missed QC hold points.
shrinkage_risk_season	Categorical	Site shrinkage risk classification: LOW, MEDIUM, or HIGH. Derived from site climate data.
wind_exposure_category	Categorical	Sheltered, Normal, or Exposed. High wind accelerates evaporation and plastic shrinkage cracking.
site_environment	Categorical	Inland, Coastal, or Industrial. Coastal / industrial environments increase chloride and carbonation risk.
accessibility	Categorical	Open, Semi-enclosed, or Enclosed. Influences wind speed and temperature at the pour surface.
city	Categorical	Construction city. Maps to climate zone for default temperature and humidity parameters.
project_tier	Categorical	Class A / B / C contractor classification based on past performance audit results.
count_similar_elements	Numeric	Number of structurally similar elements already poured on site. Used to estimate population-level crack recurrence risk.

1.3 Models & Accuracy

Three independent XGBoost pipelines — one per target. All share identical preprocessing: median imputation + StandardScaler (numeric); most-frequent imputation + OneHotEncoder (categorical).

Target	Task	Algorithm & Key Hyperparameters	Test Accuracy
crack_occurred	Binary Classification	XGBClassifier n_estimators=300, max_depth=5 learning_rate=0.05, subsample=0.85	AUC-ROC: 0.94 F1 (weighted): 0.91
crack_severity	Multi-class (4 labels)	XGBClassifier n_estimators=300, max_depth=4 learning_rate=0.05, subsample=0.85	Accuracy: 0.87 F1 (weighted): 0.86
crack_type	Multi-class (5 labels)	XGBClassifier n_estimators=300, max_depth=4 learning_rate=0.05, subsample=0.85	Accuracy: 0.83 F1 (weighted): 0.82

1.4 Prediction Outputs

Output	Classes / Range	Meaning
Crack Probability	0 – 100 %	Probability that a crack will occur. Returned by predict_proba() of the occurrence classifier.
Crack Severity	None · Minor · Moderate · Severe	'None' is shown when probability < 25 %. Driven by curing deficit, W/C excess, and placing temperature.
Crack Type	Plastic Shrinkage · Plastic Settlement · Drying Shrinkage · Thermal · Structural	Dominant crack mechanism predicted from environmental conditions and material parameters.

2. Defect Volume Intelligence Module

Predicts four outputs from 26 project, trade, material, site, and QA/QC features: (1) defects per floor, (2) dominant defect type, (3) severity grade, and (4) root cause. All cost-related fields are explicitly excluded.

2.1 Dataset Overview

Property	Detail
Total Samples	4,000 rows
Feature Count	26 input features
Targets	4 (defect_count · defect_type · severity_grade · root_cause)
Generation	Synthetic — multi-feature score-based rules tied to construction stage, QC compliance, subcontractor class, material status, SPI, and workforce skill ratio
Train / Test Split	80% / 20% (3,200 train · 800 test)
Random Seed	42

2.2 Input Features

Group A — Project & Trade Profile

Feature	Type	Description
project_type	Categorical	Residential, Commercial, Industrial, or Infrastructure. Governs baseline defect rates and inspection frequency.
gfa_sqm	Numeric	Gross Floor Area in square metres. Larger footprint means more concurrent trades and higher defect exposure.
total_floors	Numeric	Total number of floors. Used to contextualise defect count per floor.
structural_system	Categorical	RCC Frame, Shear Wall, Flat Slab, or Steel Frame. Affects which defect type dominates (Structural vs. Finishes).
subcontractor_class	Categorical	Class A / B / C based on historical performance. Class C generates 2–4× more defects per floor than Class A.
past_defect_rate_per_floor	Numeric	Historical average defects per floor from previous projects by the same subcontractor. Strongest predictor of future defect count.
workforce_size	Numeric	Total workers on site. Larger crews increase coordination complexity and supervision demand.
skill_ratio	Numeric	Fraction of workforce classified as skilled (0–1). Below 0.60 significantly increases process-related defects.
site_engineer_experience_yrs	Numeric	Years of experience of the lead site engineer. Below 4 years flags elevated supervision risk.

Group B — Activity & Progress

Feature	Type	Description
construction_stage	Categorical	Foundation, Substructure, Superstructure, MEP Rough-in, Finishes, or Facade. Primary driver of the dominant defect type.

Feature	Type	Description
spi	Numeric	Schedule Performance Index (Earned Value ÷ Planned Value). SPI < 0.90 signals schedule pressure; rushed work directly raises defect rates.
concurrent_activities	Numeric	Number of active trade packages simultaneously on site. Higher values reduce QC attention per activity.
prior_rework_same_element	Binary	1 = element has already been reworked. Prior rework approximately doubles subsequent defect probability on the same element.
qc_hold_point_compliance_pct	Numeric	Fraction of mandatory QC hold points signed off before proceeding (0–1). Below 0.80 is the primary process-defect driver in the model.
third_party_inspection	Binary	1 = client-appointed third-party inspector present. Projects with TPI typically show 30–50% lower defect escape rates.

Group C — Material & Site

Feature	Type	Description
material_grade	Categorical	Concrete or material grade: M25, M30, M35, M40. Tied to mix design compliance requirements.
approved_supplier	Binary	1 = material from a pre-qualified approved supplier. Unapproved suppliers raise material defect risk.
delivery_variance_days	Numeric	Days by which material delivery deviated from programme. Positive values indicate late delivery; negative values indicate early delivery.
test_certificate_status	Categorical	Pass, Fail, or Pending. Using materials with Fail or Pending certificates is a primary material-defect driver.
site_storage_condition	Categorical	Good, Fair, or Poor. Poor conditions degrade material properties before use.
non_productive_days	Numeric	Days lost to weather or other downtime in the reporting period. Values above 7 signal weather-driven defect risk.

Group D — Historical QA/QC

Feature	Type	Description
defects_recorded_to_date	Numeric	Running total of defects raised on this project. A rising count signals systemic process or material failure.
defect_rate_current_project	Numeric	Current project defect rate (defects per floor to date). Compared against portfolio average as a benchmark.
top_defect_type_1	Categorical	Most frequent defect category raised so far: Concrete Defects, Waterproofing, MEP Installation, Tiling/Finishing, or Structural.
defect_closure_rate_pct	Numeric	Fraction of raised defects that have been formally closed (0–1). Low values indicate backlog and systemic QC failure.
portfolio_avg_defect_rate	Numeric	Average defect rate across all active projects in the portfolio. Used as a baseline benchmark for comparison.

2.3 Models & Accuracy

Four independent XGBoost pipelines — one per target. The count pipeline uses XGBRegressor; the three classification pipelines use XGBClassifier with LabelEncoder stored on the pipeline for human-readable inference.

Target	Task	Algorithm & Key Hyperparameters	Test Accuracy
defect_count	Regression	XGBRegressor n_estimators=300, max_depth=5 learning_rate=0.05, subsample=0.85	R^2 = 0.918 MAE = 1.33 defects/floor
defect_type	Multi-class (4 labels)	XGBClassifier n_estimators=300, max_depth=4 learning_rate=0.05, subsample=0.85	Accuracy = 99.5% F1 (weighted) = 0.995
severity_grade	Multi-class (4 labels)	XGBClassifier n_estimators=300, max_depth=4 learning_rate=0.05, subsample=0.85	Accuracy = 82.5% F1 (weighted) = 0.826
root_cause	Multi-class (4 labels)	XGBClassifier n_estimators=300, max_depth=4 learning_rate=0.05, subsample=0.85	Accuracy = 88.6% F1 (weighted) = 0.884

2.4 Prediction Outputs

Output	Classes / Range	Meaning
Defect Count	0 – 50 (integer)	Predicted number of defects per floor for the next activity, given current site conditions.
Defect Type	Structural · MEP · Finishes · Facade	Dominant defect category expected. Driven primarily by construction stage and historical top defect type.
Severity Grade	Minor · Moderate · Major · Critical	Minor = cosmetic only. Moderate = rework required. Major = regulatory concern. Critical = stop-work / safety risk.
Root Cause	Process · Material · Supervision · Weather	Primary causal factor. Process = QC failures. Material = supplier or test issues. Supervision = skill or experience gaps. Weather = non-productive days.

3. Shared Pipeline Architecture

Both modules use an identical scikit-learn + XGBoost pattern ensuring reproducible, version-controlled, and auditable predictions.

Stage	Implementation	Purpose
Numeric Imputation	SimpleImputer(strategy='median')	Fills missing numeric values without introducing distribution bias.
Numeric Scaling	StandardScaler	Normalises all numeric features to zero mean and unit variance before tree induction.
Categorical Imputation	SimpleImputer(strategy='most_frequent')	Fills missing categoricals with the modal category observed during training.
Categorical Encoding	OneHotEncoder(handle_unknown='ignore')	One-hot encodes all categoricals. Unknown categories seen at inference are safely ignored.
Column Routing	ColumnTransformer	Routes numeric and categorical features to their respective preprocessing sub-pipelines in parallel.
Model	XGBRegressor or XGBClassifier (tree_method='hist')	Gradient-boosted trees. hist mode enables fast, memory-efficient training on tabular data.
Serialisation	joblib.dump / joblib.load	Preprocessor + model saved as one .joblib file for atomic, zero-drift inference.
Label Encoding	sklearn LabelEncoder stored as pipe.label_encoder_	Attached to each classification pipeline so human-readable labels are recovered at inference with inverse_transform().

4. Saved Model Files

File	Module	Target	Type
crack_occurrence_model.joblib	Crack Intelligence	crack_occurred	XGBClassifier Pipeline
crack_severity_model.joblib	Crack Intelligence	crack_severity	XGBClassifier Pipeline
crack_type_model.joblib	Crack Intelligence	crack_type	XGBClassifier Pipeline
defect_count_model.joblib	Defect Volume Intelligence	defect_count	XGBRegressor Pipeline
defect_type_model.joblib	Defect Volume Intelligence	defect_type	XGBClassifier Pipeline
defect_severity_model.joblib	Defect Volume Intelligence	severity_grade	XGBClassifier Pipeline
defect_rootcause_model.joblib	Defect Volume Intelligence	root_cause	XGBClassifier Pipeline

All .joblib files are excluded from Git via .gitignore. Regenerate locally: python generate_defect_dataset.py → python train_defect_models.py and the corresponding crack training scripts.

5. Accuracy Summary

Model File	Target	Metric	Score	Rating
crack_occurrence_model	Crack Probability	AUC-ROC	0.940	Excellent
crack_severity_model	Crack Severity	F1 (weighted)	0.860	Very Good
crack_type_model	Crack Type	F1 (weighted)	0.820	Good
defect_count_model	Defects / Floor	R^2	0.918	Excellent
defect_type_model	Defect Type	F1 (weighted)	0.995	Excellent
defect_severity_model	Severity Grade	F1 (weighted)	0.826	Good
defect_rootcause_model	Root Cause	F1 (weighted)	0.884	Very Good

All scores measured on a held-out 20% test split (random_state=42). Excellent ≥ 0.90 | Very Good 0.85–0.89 | Good 0.80–0.84.